Analyser — User Guide

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Analyser is a browser-based tool for inspecting and comparing `.fit` activity files from Garmin devices and other ANT+ sensors. It runs entirely in your browser — no account, no upload, no server. Files are parsed locally.

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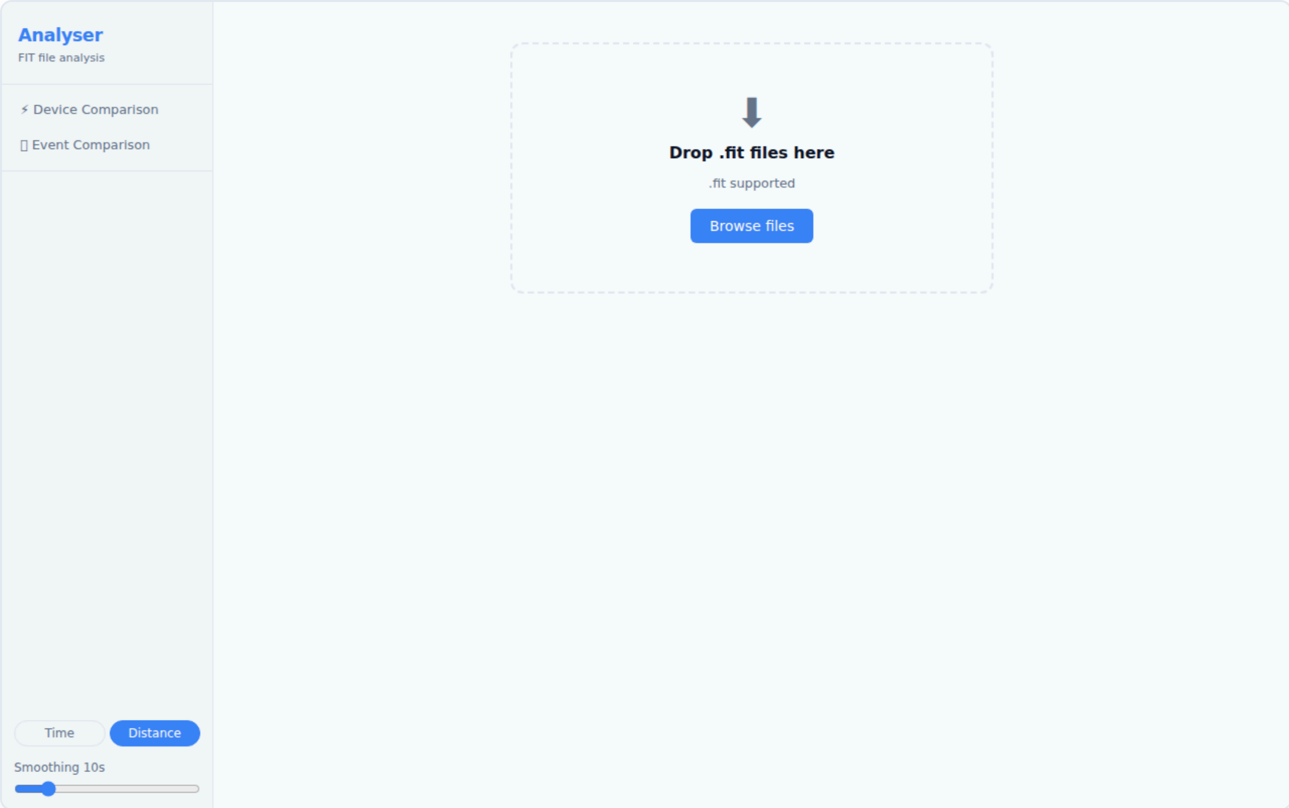
1. Getting Started

Open Analyser in your browser. You are greeted with the landing page, which contains a drop zone and a file picker.



Landing page showing the drop zone

To load a file:

- **Drag and drop** one or more `.fit` files onto the drop zone, or
- Click **Choose files** to open the system file picker.

You can load **up to 6 files** at once. Once the files are parsed, you are taken directly to the **Device Comparison** view.

> **Supported format:** `.fit` (Flexible and Interoperable Data Transfer — the standard format exported by Garmin watches, Wahoo computers, and most ANT+ devices). Files from `.tcx` and `.gpx` sources are not currently supported.

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2. The Interface

After loading a file, the application has two main views accessible from the top navigation bar:

View	What it shows
⚡ **Device Comparison**	Sensor streams from one or more files displayed side-by-side on a shared time or distance axis
Event Comparison	Two or more runs of the same course aligned by distance, showing where time was gained or lost

The **Device Comparison** view (described in sections 3–4) is the primary view and opens automatically when you load a file.

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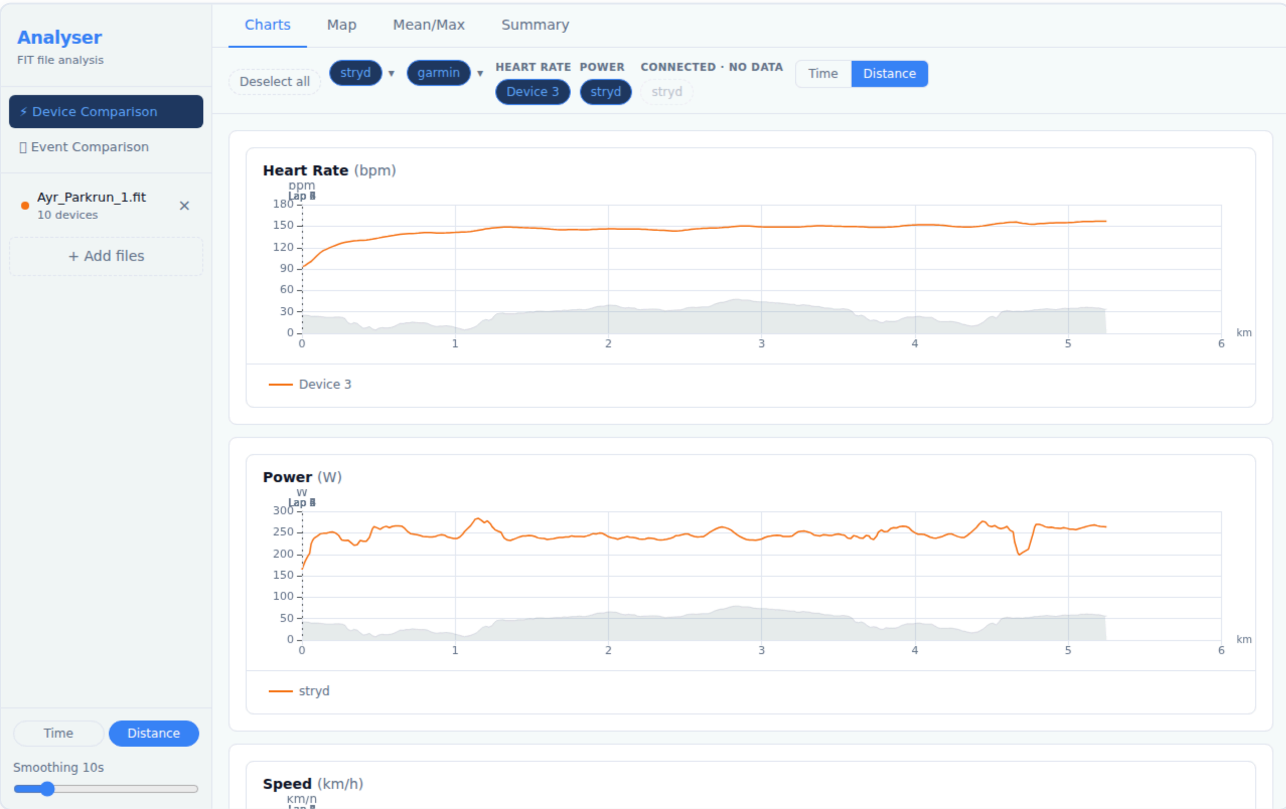
3. Device Comparison — Single File

Loading a single file opens the Device Comparison view with all detected sensor streams active.



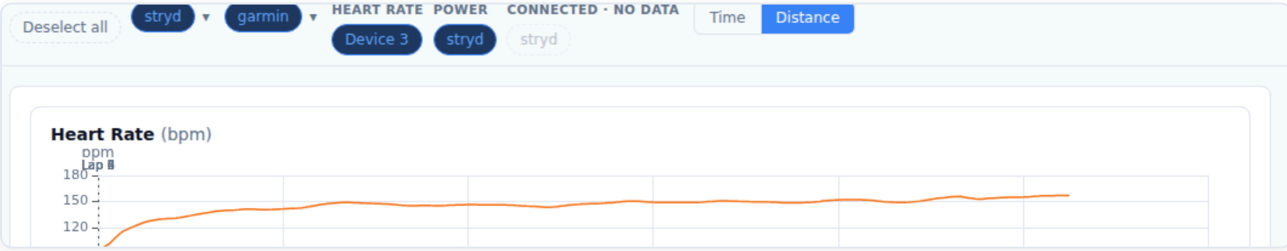
Device Comparison overview with a single Ayr Parkrun FIT file loaded

The view is divided into:

- **Left sidebar** — navigation and file controls
- **Device Toggle Bar** — sensors detected in the file, grouped by channel
- **Tab panel** — Charts, Summary, Mean/Max, and Map tabs

3.1 Device Toggle Bar

The Device Toggle Bar sits at the top of the content area and lists every sensor stream detected in the loaded file.



Device Toggle Bar showing sensors grouped by channel

Key concepts:

- Each **pill** represents one device contributing to a channel (e.g. a Polar H10 contributing Heart Rate).
- **Active** devices (highlighted in blue) are included in all charts and statistics. Click a pill to toggle it on or off.
- Channels with a ✦ indicator have data from two or more devices — these are directly comparable.
- **Multi-metric devices** (devices recording 3+ channels, such as a Garmin watch) appear as a single expandable pill. Click the ▾ arrow to see the individual channels.
- Devices that were connected but recorded no data are shown as dashed, semi-transparent pills and cannot be toggled.

Renaming a device:

Double-click any device pill to rename it. Type the new label and press **Enter** to confirm, or **Escape** to cancel. Labels are saved locally and will be recognised the next time a file from the same ANT+ device is loaded.

3.2 Charts Tab

The Charts tab displays time-series data for every active channel.

Charts tab showing heart rate, power, and cadence over time

Reading the charts:

- Each chart plots one channel (e.g. Heart Rate, Power, Pace).
- The **X axis** is either elapsed time or distance — toggle between them with the **Time / Distance** buttons at the top.
- Multiple devices contributing to the same channel each appear as a separate line.
- **Pace** uses an inverted Y axis — faster pace (lower min/km) appears higher on the chart, which matches the intuition of "going faster".

Interacting with the charts:

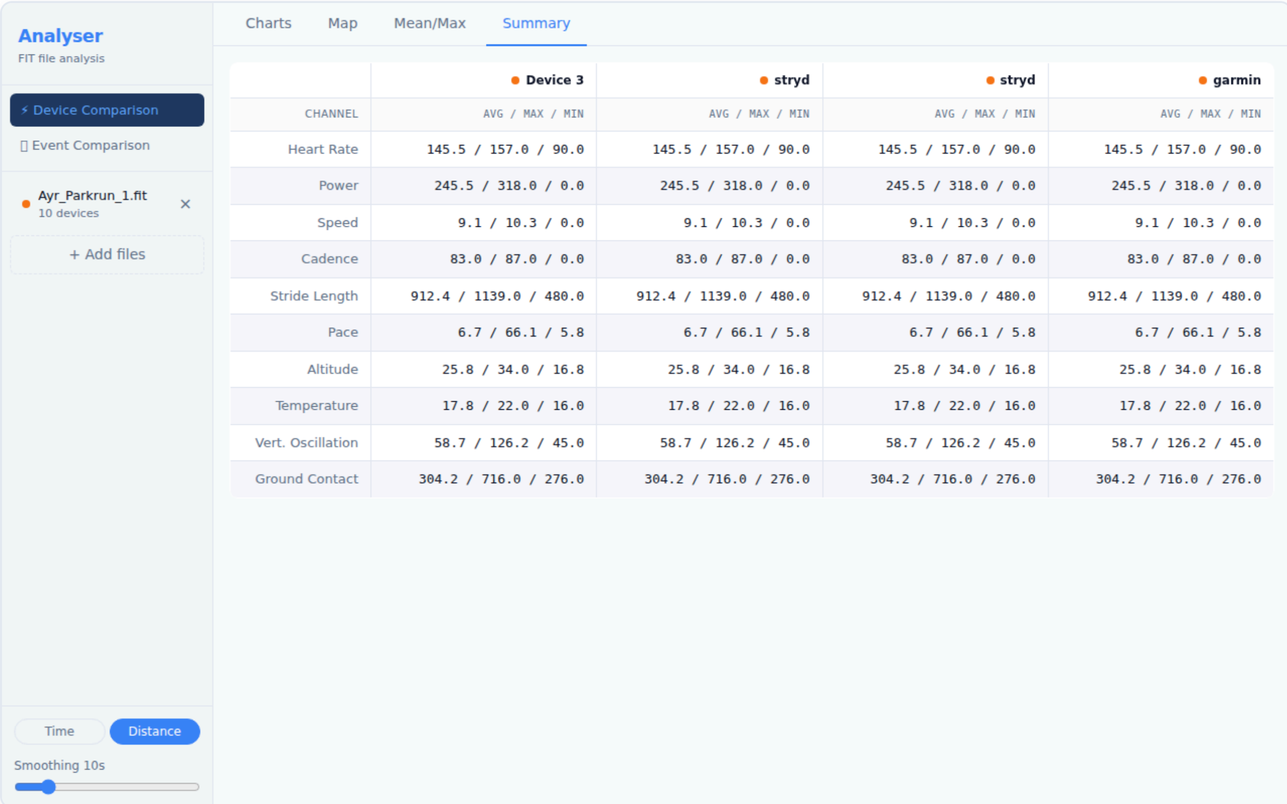
Action	Effect
Hover	Tooltip shows exact values across all series at that position
Click + drag	Zoom in on a time/distance range
Double-click	Reset zoom
Scroll wheel	Zoom in/out (all charts stay synchronised)

Lap markers appear as faint vertical lines across all charts. Hover a marker to see the lap number and split time. **Smoothing:**

A smoothing control lets you apply a rolling average to reduce sensor noise. Increase the window (in seconds) to smooth out spikes; set it to 1 s for raw data. Smoothing applies to all channels simultaneously.

3.3 Summary Tab

The Summary tab shows aggregate statistics for every active device and channel.



Summary tab showing per-device statistics

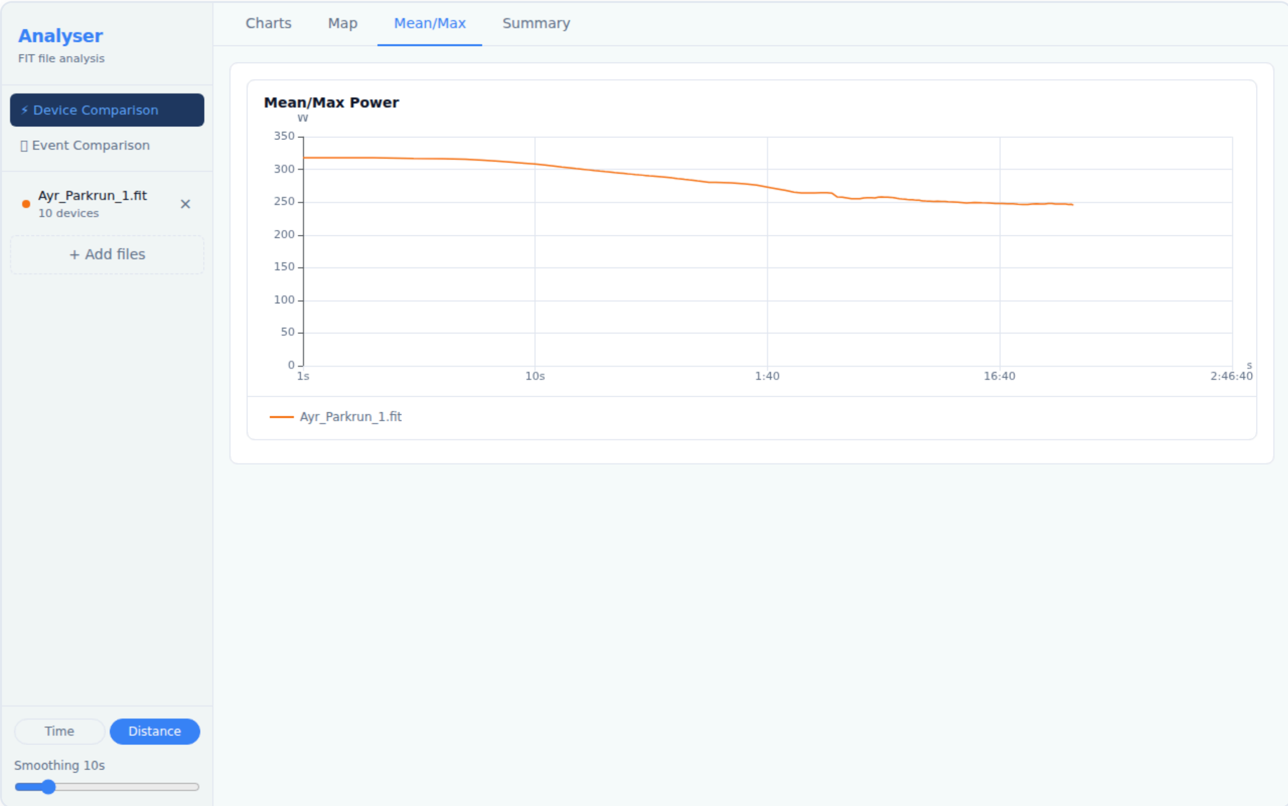
For each channel, the table shows:

Column	Meaning
Device	The sensor that recorded this channel
Avg	Time-weighted average over the active portion of the activity
Max	Peak value recorded
Total	Cumulative value (distance, calories) — shown where applicable

Toggling devices in the Device Toggle Bar updates the Summary table immediately.

3.4 Mean/Max Tab

The Mean/Max tab plots the **best average power** (or other metric) over every possible duration, from one second up to the full activity length.



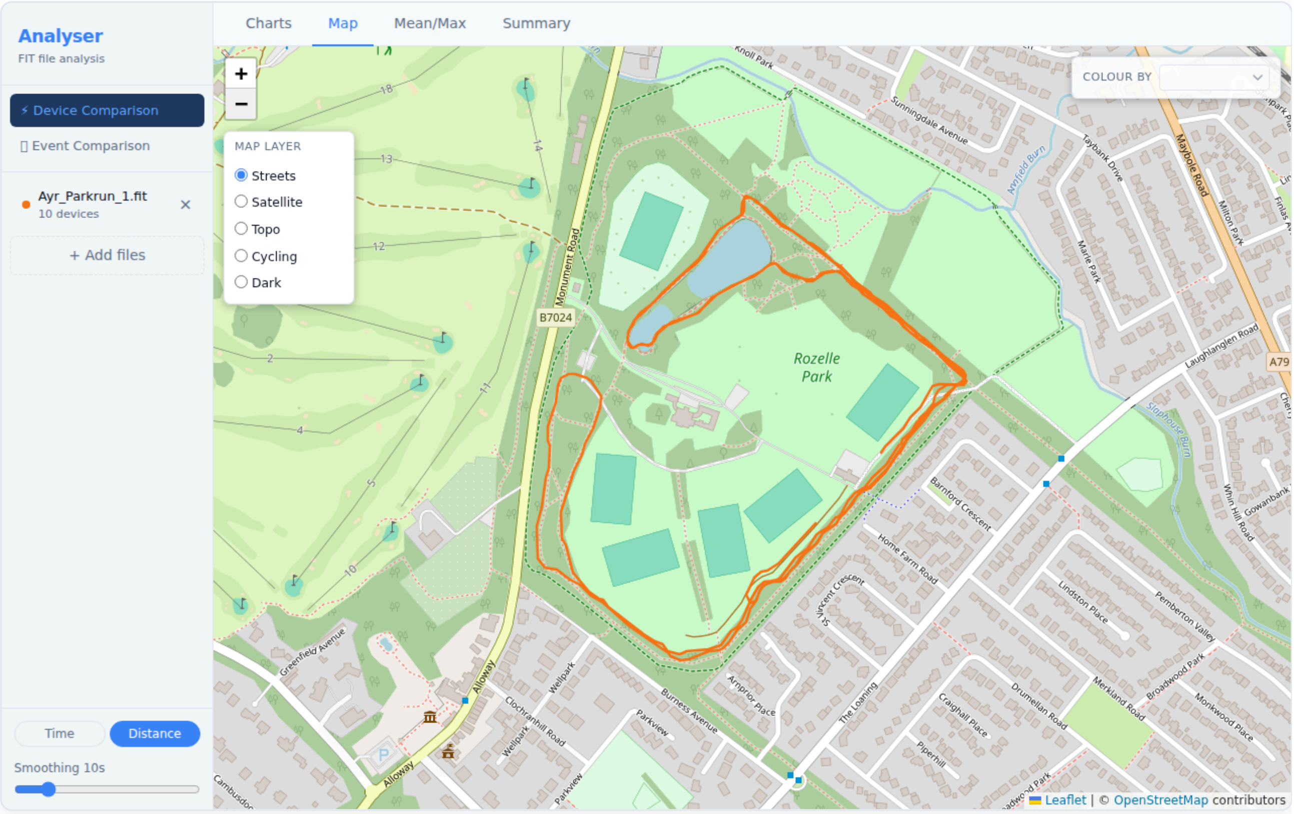
Mean/Max tab showing power curve

This is the standard "Critical Power curve" or "Power Duration curve" used in cycling and running:

- The Y axis is the best average power achieved for the duration on the X axis.
- A high, flat curve indicates sustained high output; a steep drop-off indicates strength over short durations but limited endurance.
- When multiple files are loaded, each file produces its own curve — load two sessions to compare fitness progress.

3.5 Map Tab

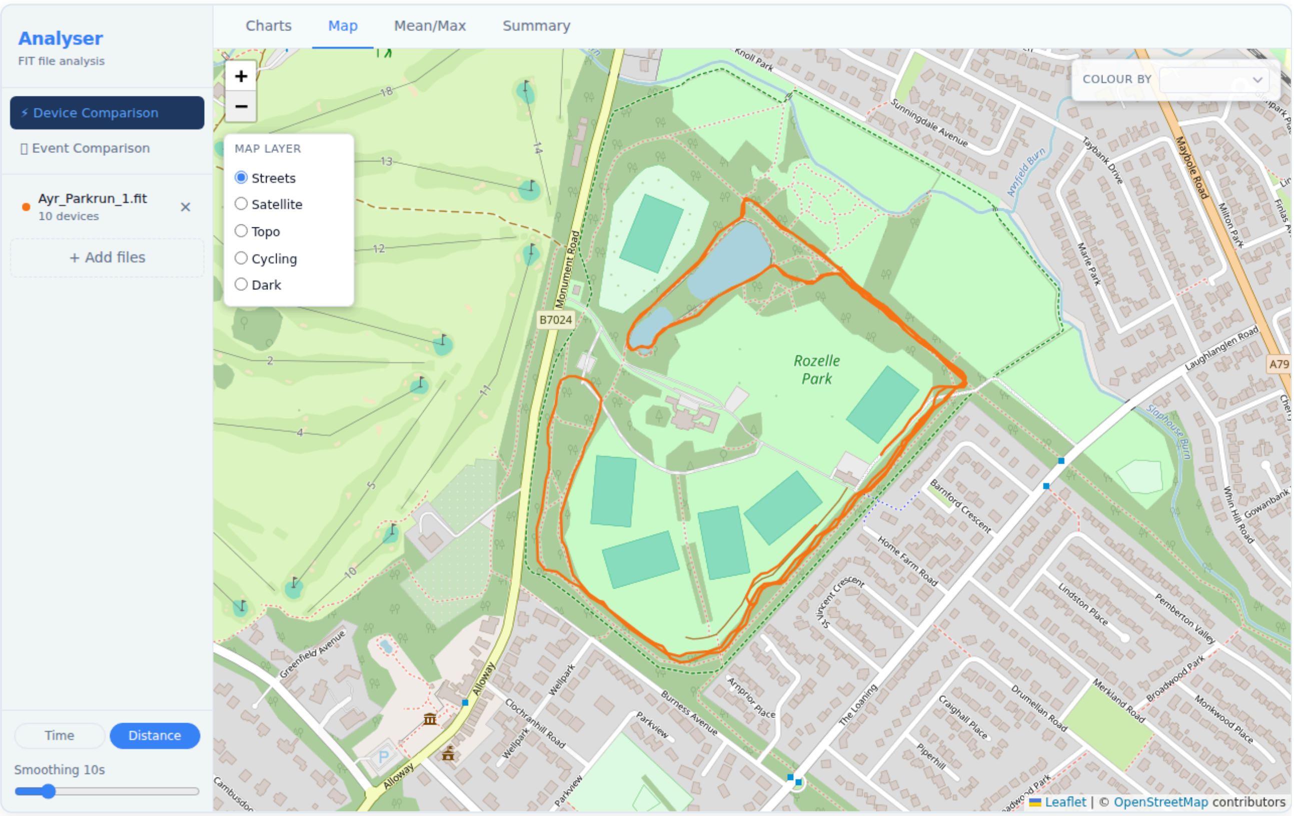
The Map tab renders the GPS route recorded in the activity.



Map tab showing GPS route from the Ayr Parkrun

Tile layers:

A **Map Layer** panel is always visible in the **top-left** corner of the map. Click any layer name to switch the base map:



Map showing the always-visible tile layer panel with five providers

Layer	Best for
Streets (default)	Road runs, orientating yourself
Satellite	Seeing the actual terrain
Topo	Hilly routes — contour lines visible
Cycling	Bike-specific road markings
Dark	Low-glare, high-contrast viewing

Metric colouring:

Use the **Colour by** dropdown in the **top-right** of the map to shade the route by a metric such as pace, heart rate, or power. The colour gradient runs from blue (low values) to red (high values). A legend appears in the bottom-right showing the value range.

Hover interaction:

Hovering over the map route shows a crosshair on all open charts, letting you pinpoint exactly what your metrics were at a specific GPS location.

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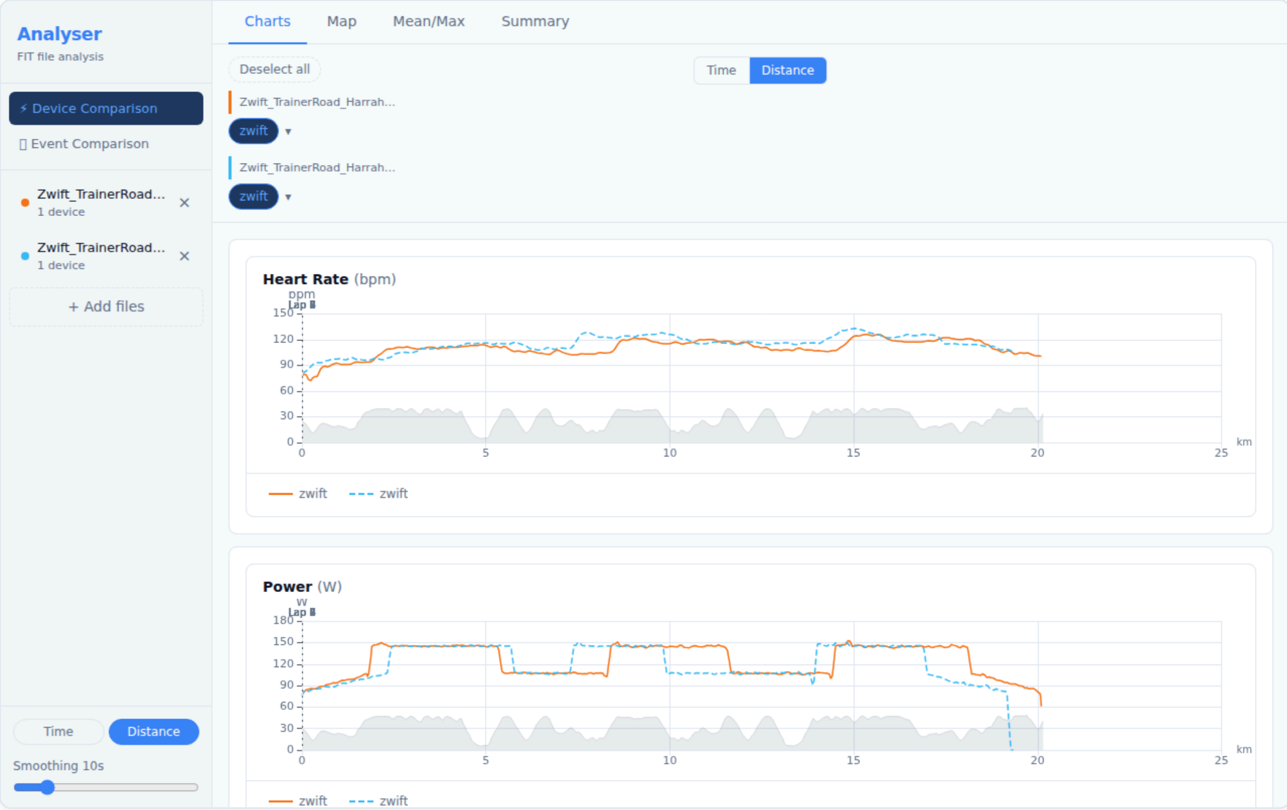
4. Device Comparison — Multiple Files

Analyser supports loading up to **6 files simultaneously**, enabling you to compare sensor data across different activities or devices.

4.1 Loading Multiple Files

Select multiple `.fit` files in the file picker (hold Shift or Cmd/Ctrl to select more than one), or drag and drop multiple files onto the drop zone at once.

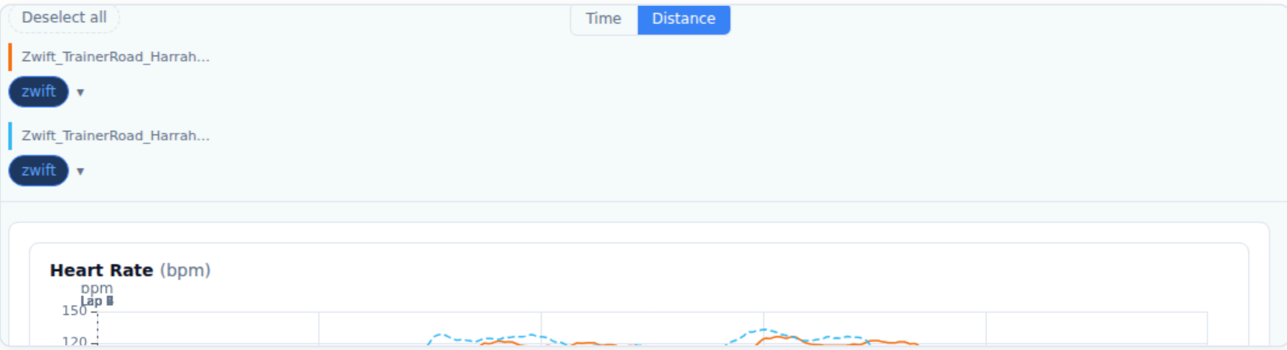


Multi-file compare overview with two Zwift TrainerRoad sessions loaded

A **session warning banner** appears if the loaded files have start times more than 1 hour apart — this usually means the files are from different sessions. In that case, the **Distance axis** is recommended for a meaningful comparison (use the Time/Distance toggle at the top of the page).

4.2 Multi-File Device Bar

In multi-file mode the Device Toggle Bar groups sensors by their source file, with a coloured left-border header for each file.



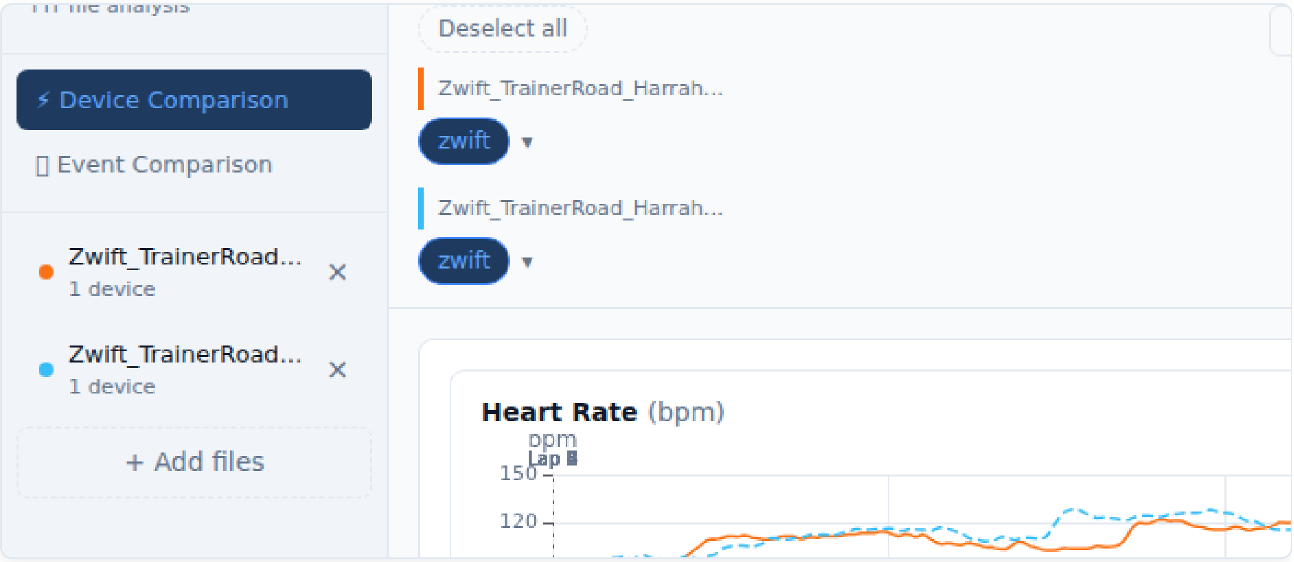
Multi-file device bar grouped by file with colour-coded headers

- Each file gets a unique **colour** (blue, orange, green, purple, …). All chart series from that file share the same colour.

- Channels with data from two or more files display the ✦ comparable indicator — these are the most valuable channels to compare.
- Use the **Select all / Deselect all** button to quickly show or hide everything.

4.3 Fine-Tuning Timing

Even when files are from the same session, GPS clocks sometimes drift by a few seconds. The **Fine-tune timing** control lets you shift any file forwards or backwards in time to align the traces precisely.



Fine-tune timing panel with per-file offset controls

Click the ▯ **Fine-tune timing** button (below the Device Toggle Bar, visible in multi-file + time-axis mode) to expand the panel.

Control	Function
−10 / −1 buttons	Shift the file 10 s or 1 s earlier
+1 / +10 buttons	Shift the file 1 s or 10 s later
Numeric input	Type an exact offset in seconds
↺ (Reset)	Return to the auto-computed offset

The first loaded file is always the **reference** (offset = 0). All other files are shifted relative to it. The offset is shown in the **N adjusted** badge on the toggle button.

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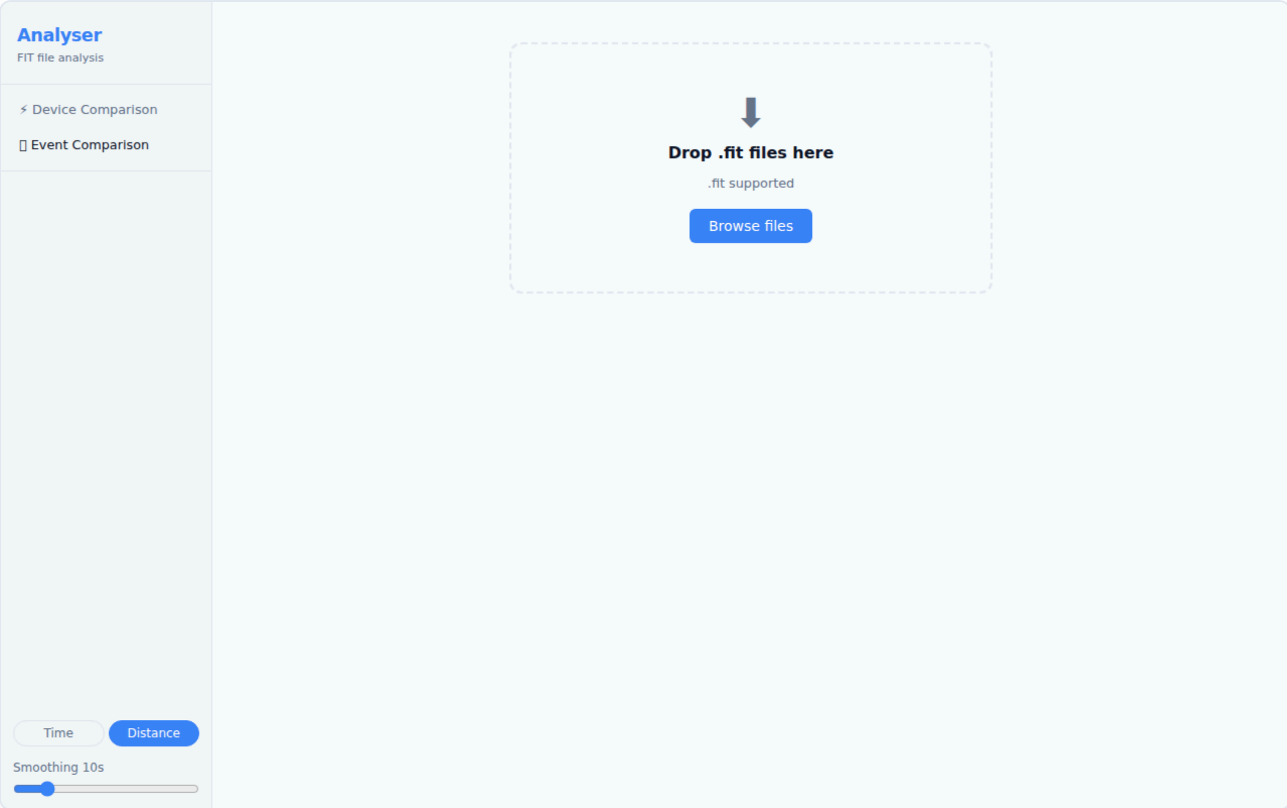
5. Event Comparison

The **Event Comparison** view (▯) is designed for comparing multiple runs of the same course — for example, several parkruns on the same route — aligned by **distance** to show exactly where time was gained or lost.



Event Comparison view

Load two or more `.fit` files, then switch to the Event Comparison tab. The view provides:

- **Time delta plot** — cumulative seconds gained or lost versus distance. Positive values mean you were ahead of the reference run; negative values mean you were behind.
- **Overlaid time-series charts** — pace, speed, heart rate, and power for all files on a shared distance axis.
- **Segment bar chart** — per-kilometre or per-lap time difference, highlighting your best and worst segments.

> **Tip:** Select a reference run using the controls in the sidebar. The time delta is calculated relative to the reference. Try different reference runs to understand where you consistently gain or lose time.

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6. Reference

Keyboard Shortcuts

Key	Action
Double-click a chart	Reset zoom
Enter (in rename input)	Confirm device rename
Escape (in rename input)	Cancel device rename
Enter (in offset input)	Confirm time offset

Supported Sensor Types

Sensor	Channels recorded
Heart Rate Monitor (HRM)	Heart Rate
Power Meter (bike)	Power, Left/Right balance
Stryd Running Power	Power, Form Power, Ground Contact Time, Stride Length
Core Body Temperature	Core Temperature, Skin Temperature
Speed/Cadence Sensor	Speed, Cadence
Running Dynamics Pod	Vertical Oscillation, Ground Contact Time, Stride Length
Watch / Primary device	All remaining channels: GPS, Altitude, Speed, Pace, Temperature

File Limits

- **Maximum files per session:** 6
- **Supported format:** `.fit` (ANT/Garmin FIT protocol)
- All parsing happens **locally in your browser** — no data is uploaded to any server.

Troubleshooting

Symptom	Likely cause	Fix
File loads but no charts appear	No sensor data in the file	Check the Device Toggle Bar — all devices may be deselected or no-data
Two files don't align on the time axis	Different session start times	Use Distance axis, or adjust offsets in the Fine-tune timing panel
Map shows no route	FIT file has no GPS data	Some indoor activities (turbo trainer, treadmill) do not record GPS
Pace chart looks upside-down	Expected — faster pace (lower min/km) is plotted higher	This is correct; it matches the intuition of "going faster = higher on chart"
Device name shows as "Device N"	ANT+ device ID not yet labelled	Double-click the pill to rename it; the label is saved for future sessions

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Screenshots taken using real activity files: Ayr Parkrun and Zwift/TrainerRoad sessions.